Software Requirements Specification

Version 1.0

IoT-Integrated Smart Warehouse & Inventory Logistics Portal

Project #22

CS 251 — Software Engineering

2. Functional Requirements

Functional requirements define the specific behaviors, actions, and operations the system must support. Each requirement is expressed as a testable system obligation.

2.1 Inventory & Storage Management

FR1: The system shall suggest storage locations for items based on sales velocity and physical dimensions (height, width, depth).

FR2: The system shall calculate the 3D volumetric usage of each zone and aisle, and shall reject item assignment requests that would cause a bin to exceed its defined capacity.

FR3: The system shall continuously compare shelf IoT sensor weight readings against the expected weight derived from the current inventory count, and shall raise a discrepancy alert when the deviation exceeds a configurable tolerance threshold.

FR4: The system shall flag items whose Best Before date falls within a configurable warning window and shall recommend First-Expiry, First-Out (FEFO) picking order for those items.

FR5: The system shall monitor simulated environmental conditions (temperature and humidity) per storage zone and shall trigger an alert when items classified as Sensitive are stored in a non-compliant zone.

FR6: The system shall enforce a soft-lock on inventory items added to a customer cart, marking their state as Reserved and excluding them from the available stock count until the cart is either confirmed or abandoned.

FR7: The system shall automatically schedule aisles for cycle count audits based on the elapsed time since the last audit or based on a historically high discrepancy rate for that aisle.

FR8: The system shall identify incoming shipment items that fulfill existing open backorders and shall route those items directly to the packing station, bypassing standard storage intake.

FR9: The system shall monitor total warehouse occupancy and shall notify the Warehouse Manager when occupancy exceeds 90% of total capacity.

FR10: The system shall transition inbound shelf goods through the following receiving state machine, enforcing sequential order with no skipped or reversed transitions:

Expected → At Dock → Being Inspected → Stored

2.2 Supplier & Procurement

FR11: The system shall monitor stock levels against each SKU's defined safety stock threshold and shall automatically notify the Supplier module when stock reaches or falls below that threshold (Observer Pattern).

FR12: The system shall compute an estimated shipment arrival date for each active purchase order based on the responsible supplier's historical delivery performance data.

FR13: The system shall auto-generate a structured Purchase Order (PO) containing itemized SKUs, unit prices, quantities, total cost, and a simulated digital signature upon procurement trigger.

FR14: The system shall calculate and maintain a supplier accuracy score, defined as the ratio of correctly received items to the total items ordered, updated after every inbound shipment reconciliation.

FR15: The system shall automatically select between a primary and a secondary supplier for a given SKU based on current unit price and estimated delivery speed, according to a configurable selection rule.

FR16: The system shall store and version all supplier contracts, including multiple pricing tiers and volume discount thresholds, and shall retain historical versions for auditing.

FR17: The system shall prioritize the allocation of incoming stock to customers with the oldest open backorder dates first, before making stock available to new orders.

2.2 Order Fulfillment & Shipping

FR18: The system shall group multiple open orders into a single optimized pick list and shall compute a walking route for floor staff that minimizes total travel distance across aisles.

FR19: The system shall provide a sort-to-light guidance interface that directs a packer to assign each picked item to the correct customer-specific shipping bin, requiring no more than two interactions per item.

FR20: The system shall compare the final packed parcel weight against the sum of the expected weights of all constituent items and shall flag any deviation exceeding a configurable tolerance as a packing error.

FR21: The system shall select a shipping carrier for each order based on the customer's delivery address and a configurable preference rule (e.g., fastest delivery versus lowest cost).

FR22: The system shall transition each order through the following fulfillment state machine, enforcing sequential order:

Processing → Picking → Packing → Shipped → Delivered

FR23: The system shall recommend the smallest viable box size for a given order based on the combined dimensions of all items in the order, to minimize shipping cost and material waste.

FR24: The system shall generate a unique, scannable shipping label for each fulfilled order that encodes a QR code linking the physical parcel to its corresponding digital order record.

2.3 Access, Security & Administration

FR25: The system shall enforce role-based access control (RBAC), restricting Floor Staff accounts to picking and packing screens, and granting Warehouse Manager accounts exclusive access to procurement, analytics, and administrative modules.

FR26: The system shall provide an Emergency Mode that, upon a single authenticated action by an authorized manager, immediately pauses all active picking tasks and locks all dock doors. The activation event shall be persisted to the audit log with a timestamp, the triggering user's identity, and a mandatory reason field.

FR27: The system shall automatically archive completed order records to a long-term read-only storage tier after 12 months of inactivity, without degrading performance of queries on active records.

FR28: Multi-Warehouse Data Aggregator Allowing a manager to view "Global Stock" across multiple physical fulfillment centers.

FR29: System Audit Trail A permanent log of every "Inventory Adjustment" (e.g. manually changing a stock count) with a required reason-code.

Fr30: Pick-Failure Protocol A workflow for when an item is missing from its shelf, triggering an immediate "In ventory Audit" and re-routing the picker.

3. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints the system must satisfy. They are organized by quality characteristic.

3.1 Performance

NFR1: Stock level monitoring shall detect a safety stock threshold breach and deliver the corresponding reorder trigger notification to the Supplier module within **5 seconds** of the breach event.

NFR2: Batch pick route computation shall complete and deliver an optimized pick list to the floor staff device within **3 seconds**, regardless of the number of orders batched.

NFR3: Warehouse occupancy alerts shall be raised in real time — within the same transaction cycle in which a stock assignment is committed to the database — with no perceptible delay.

3.2 Reliability

NFR4: The soft-lock (Reserved state) applied to cart items shall be enforced consistently across all concurrent user sessions with no race conditions, ensuring no item is simultaneously reserved by more than one customer.

NFR5: Both the fulfillment state machine (FR22) and the inbound receiving state machine (FR10) shall reject any transition that does not follow the defined sequential order under all possible input conditions, including concurrent requests and edge-case inputs.

NFR6: IoT weight sensor discrepancy detection shall produce zero false negatives when a weight deviation meets or exceeds the configured tolerance threshold (if there really is a discrepancy, the system must **never miss it** and must **always flag it**.).

3.3 Scalability

NFR7: The archive logic (FR27) shall maintain active database query performance as the completed orders table grows.

NFR8: The cycle count scheduler (FR7) shall operate correctly on warehouses of arbitrary size — in terms of the number of aisles and zones — without requiring manual reconfiguration.

3.4 Security

NFR9: RBAC enforcement (FR25) shall be applied and verified server-side on every request; client-side role filtering alone is not a sufficient security control and shall not serve as the sole enforcement mechanism.

NFR10: Emergency mode activation shall be logged with a timestamp, the triggering user's identity, and a reason field.

NFR11: Supplier contracts, pricing tiers, and Purchase Order data shall be accessible only to users authenticated with procurement-level permissions or above; any access attempt by a lower-privileged role shall be rejected with an appropriate authorization error.

3.5 Usability

NFR12: The sort-to-light packing interface (FR19) shall require no more than **two user interactions** per item to confirm correct bin placement, minimizing cognitive load on floor staff during high-throughput packing operations.

NFR13: Emergency Mode (FR26) shall be activatable within a **single authenticated action** with no multi-step confirmation dialog, to ensure the fastest possible response during a security or safety incident.

3.6 Maintainability

NFR14: All Observer Pattern implementations — including the stock reorder trigger (FR11) and the occupancy alert (FR9) — shall be architected such that adding a new subscriber requires no modification to publisher logic, in compliance with the Open/Closed Principle.

NFR15: The supplier selection logic (FR15) shall be driven by an externally configurable rule set (e.g., a configuration file or database-stored rules), such that selection criteria can be updated without requiring code changes or application redeployment.

3.7 Data Integrity

NFR16: Archived order records (FR27) shall remain queryable in read-only mode at all times and shall never be permanently deleted from the system, in support of regulatory and audit obligations.

NFR17: All weight sensor readings used for discrepancy detection shall be stored with a tamper-evident timestamp before any downstream processing, ensuring a complete and auditable chain of sensor evidence.